Practice — Saturation, Concentration and Solubility

Section A is interactive — tap an option or type and press Check. Sections B and C are written practice.

Objective (1 mark each)

Q1 · MCQ **[1]**

A solution that can still dissolve more solute is —

(a) saturated (b) unsaturated (c) concentrated (d) solid

Q2 · MCQ **[1]**

The maximum amount of solute that dissolves in a fixed amount of solvent is its —

(a) concentration (b) solubility (c) dilution (d) saturation point

Q3 · MCQ **[1]**

For most solids, raising the temperature makes solubility —

(a) decrease (b) increase (c) stay the same (d) become zero

Q4 · MCQ **[1]**

A solution with a large amount of solute is called —

(a) dilute (b) concentrated (c) unsaturated (d) pure

Q5 · TRUE / FALSE **[1]**

A saturated solution can dissolve more solute if it is heated.

(a) True (b) False

Q6 · FILL IN THE BLANK **[1]**

A solution in which no more solute can dissolve at a given temperature is ________.

____

Q7 · FILL IN THE BLANK **[1]**

The amount of solute in a fixed quantity of solution is its ________.

____

Short answer (2 marks each)

Q8 **[2]**

Differentiate between a saturated and an unsaturated solution.

Q9 **[2]**

Define solubility and state how temperature affects the solubility of most solids.

Q10 **[2]**

Which is more concentrated: 3 spoons of sugar in 200 mL or 3 spoons in 100 mL? Why?

Long / application (3 marks each)

Q11 **[3]**

Describe an activity to show that solubility of a solid increases with temperature.

Q12 **[3]**

Your friend says "a saturated solution can never dissolve more solute." Correct this statement.

Check yourself → Answer key